The Indus civilisation: its history, and what is true about its language

Arham Partap Jain's Indus-script project (github.com/Savirinc/indus-script). Compiled 29 October 2026. A reference record, kept separate from opinion. Every statement carries a status label and a source.

How to read this document

Label	Meaning
ESTABLISHED	Accepted by specialists of every school. Based on excavation, dated finds or the texts themselves.
STRONG EVIDENCE	Supported by more than one independent line of evidence, but disputed by some, or resting on interpretation.
PROPOSED	A hypothesis argued in print. Plausible, not demonstrated.
UNKNOWN	No evidence can settle it at present. Stated so as not to be mistaken for knowledge.

The short answer. We know a great deal about how the Indus people lived. We know almost nothing certain about the language they spoke. It existed, and it was foreign enough to Mesopotamians to need an interpreter ESTABLISHED. It was probably not Indo-Aryan (Sanskrit's ancestor) STRONG EVIDENCE. Dravidian, Munda-related or a lost language are all PROPOSED. The script is undeciphered, and no sign has an established sound or meaning ESTABLISHED. What would change this is outside evidence that ties signs to sounds: a bilingual text, or names that can be matched to particular seals.

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1. Discovery

- Alexander Cunningham published the first Indus seal, found at Harappa, in 1875 (*Archaeological Survey of India Report* 5). He did not know how old it was. ESTABLISHED
- Daya Ram Sahni excavated Harappa (Punjab, now Pakistan) in 1921. R. D. Banerji began excavating Mohenjo-daro (Sindh) in 1922. ESTABLISHED
- John Marshall, Director-General of the Archaeological Survey, announced a new Bronze Age civilisation in the *Illustrated London News* on 20 September 1924. Mesopotamian scholars soon matched the seals to ones found at Susa and in Mesopotamian sites, which dated the civilisation to the 3rd millennium BCE. ESTABLISHED
- It is called the Indus or Harappan civilisation, after its first excavated city. "Indus-Sarasvati" is used by some; the name carries a claim about the Ghaggar-Hakra river (section 5). ESTABLISHED

2. Timeline

Dates are approximate and vary by author by a century or so. The phases follow Kenoyer (1998) and Possehl (2002).

Phase	Dates (BCE)	What happens
Early food-producing (Mehrgarh)	c. 7000–5500	Farming villages in Balochistan: wheat, barley, cattle, goats, mud-brick houses. Mehrgarh is one of the earliest farming sites in South Asia. ESTABLISHED
Regionalisation (Early Harappan: Hakra, Ravi, Kot Diji, Amri)	c. 5500–2600	Regional cultures spread across the plains, with walled towns and long-distance exchange. Marks on pottery at Harappa from about 3300 BCE resemble some later signs. STRONG EVIDENCE that they are precursors; whether they are writing is debated.
Mature Harappan (Integration)	**c. 2600–1900**	Planned cities, standard weights, standardised seals and **the script**, and trade with Mesopotamia. Nearly all inscriptions belong here. ESTABLISHED
Late Harappan (Localisation)	c. 1900–1300	The cities shrink or are abandoned, and settlement shifts east and south. Seals, writing and standard weights largely disappear. ESTABLISHED
Rigveda composed	c. 1500–1200	The oldest Indo-Aryan text, composed in the Punjab. It names the river *Sindhu*. It shows no knowledge of living Harappan cities. STRONG EVIDENCE (the dating is by linguistic and indirect arguments)
Next Indian writing (Brahmi, Kharosthi)	c. 3rd century BCE (Ashoka); possibly somewhat earlier	There is a gap of about 1,500 years with no known writing in South Asia. ESTABLISHED (no continuous script tradition is documented across the gap)

3. Extent, cities and way of life

- **Area.** The largest of the three early river civilisations (with Egypt and Mesopotamia), covering roughly 800,000–1,000,000 km². It ran from Balochistan and the Makran coast to Gujarat and the upper

Ganges–Yamuna plain, with outposts such as Shortugai on the Amu Darya in Afghanistan. More than a thousand Mature Harappan settlements are known. ESTABLISHED

- **Major sites:** Mohenjo-daro and Harappa, the two largest excavated cities, each well over 100 ha; Dholavira (Kutch); Ganweriwala (Cholistan, unexcavated); Rakhigarhi (Haryana); Lothal (Gujarat, with a large basin often read as a dock); Kalibangan (Rajasthan). ESTABLISHED
- **Planning.** Cities have grid-like streets, standardised baked bricks in the ratio 1:2:4, covered drains, wells and private bathing areas. Mohenjo-daro has a raised citadel mound with the "Great Bath". ESTABLISHED
- **No palaces or royal tombs** have been identified, and no monuments of kings. How the society was governed is UNKNOWN. Proposals range from merchant elites to city-states to a loose network.
- **Weights.** Cubical chert weights follow one standard throughout. The smallest common unit is about 0.87 g, and the ratios are binary at first (1, 2, 4, 8, 16, 32, 64) and decimal above. ESTABLISHED
- **Crafts.** Long carnelian beads drilled with special stone drills, at Chanhu-daro, Lothal and elsewhere; shell work; faience; stoneware bangles; copper and bronze tools; cotton cloth (among the earliest known). ESTABLISHED
- **Food.** Wheat, barley, pulses, sesame and later millets and some rice; cattle, including zebu, plus sheep, goats and buffalo. ESTABLISHED
- **Seals.** Square steatite stamp seals, usually 2–3 cm across and fired hard. They usually show an animal, most often the one-horned "unicorn", under a short line of signs. They were used to stamp clay sealings on goods and doors. ESTABLISHED
- **Religion.** Female figurines, a horned seated figure (the "proto-Shiva" seal, M-304), fire altars at Kalibangan and Lothal, and the Great Bath. All interpretations are PROPOSED. Nothing written explains them.
- **People (ancient DNA).** The one published genome of a Harappan city burial (a woman from Rakhigarhi, mid-3rd millennium BCE; Shinde et al. 2019) shows ancestry mixed from early Iranian-related and South Asian hunter-gatherer sources. It has **no steppe ancestry**. Steppe ancestry enters South Asia in the 2nd millennium BCE (Narasimhan et al. 2019). STRONG EVIDENCE (one genome from the city, plus about a dozen related individuals buried in Iran and Turkmenistan)

4. Trade with Mesopotamia and the Gulf ("Meluhha")

- **Meluhha** is the name Sumerian and Akkadian texts give a land reached by sea beyond Magan (Oman) and Dilmun (Bahrain). Most specialists identify 3rd- and early-2nd-millennium Meluhha with the Indus region. STRONG EVIDENCE (the goods, the direction and Indus objects found in Mesopotamia fit; no text names an Indus city)
- Sargon of Akkad (c. 2334–2279 BCE) boasts that ships from Meluhha, Magan and Dilmun moored at the quay of his capital, Agade. ESTABLISHED (the text)
- Gudea of Lagash (c. 2100 BCE) records materials from Meluhha, among them carnelian, wood and gold dust. Texts of the Ur III period mention a "Meluhha village" near Lagash, where people of Meluhhan origin lived. ESTABLISHED (the texts)
- **An interpreter.** The Akkadian-period cylinder seal of Shu-ilishu (Louvre) calls him "interpreter of the Meluhhan language". This is the single most important fact about the language: **Meluhhans spoke a language of their own, different from Sumerian and Akkadian, and it needed translating.** ESTABLISHED
- **Goods named** include carnelian, lapis lazuli, gold, copper, ivory, fine woods (Sumerian *mes*, *ab-ba*), a bird of Meluhha (*dar-mušen*, perhaps a peacock or a jungle fowl), dogs, boats and furniture "of

Meluhhan style". Mesopotamian scribes named these goods **in their own words**. Possehl (1996) catalogues 76 attestations of the name Meluhha, and not one is a word in the Meluhhan language. ESTABLISHED (this project, round 51)

- **Objects that moved:** Indus seals and carnelian beads have been found at Ur, Kish, Susa and in the Gulf. "Gulf" round seals from Bahrain and Failaka use Indus signs in an un-Indus order. ESTABLISHED (this project, rounds 30–31: the order is foreign, percentile 0.916)
- **The name Meluhha itself.** Its etymology is PROPOSED: Dravidian *mēl-akam*, "high country" (Parpola), or the source of later Sanskrit *mleccha*, "foreigner, barbarian" (Pali *milakkha*; Parpola, Witzel). If the *mleccha* link is right, Vedic speakers called the Meluhhans foreigners, which counts against the Harappans having spoken Sanskrit. The sound changes needed are not explained, so it stays unproven.
- **Sindhu in cuneiform.** The Akkadian *sindû*, "Indian (wood)" (GIŠ *si-in-da-a*), is a place name carried over from *Sindhu*. It is attested only in Sennacherib's inscriptions (704–681 BCE), about 1,200 years after the Harappan cities (CAD S 284). It is not Harappan-era evidence. ESTABLISHED (this project, round 51 addendum)
- By the 1st millennium BCE, "Meluhha" in Mesopotamian texts refers to Nubia and Ethiopia, not India. ESTABLISHED

5. The decline

- Between about 1900 and 1700 BCE the large cities were reduced or abandoned, and writing, seals and standard weights went out of use. Smaller farming settlements continued and spread east into the Ganges–Yamuna area. ESTABLISHED
- Causes: weakening monsoon rains (the "4.2 ka" drying event), shifting and drying rivers, and the collapse of the Gulf trade. Most specialists combine these. STRONG EVIDENCE for climate and river change (Giosan et al. 2012). How much each counted is debated.
- The Ghaggar-Hakra, which some identify with the Vedic Sarasvati, was not a large glacier-fed river in Harappan times, according to sediment studies (Giosan et al. 2012). It was monsoon-fed and dried in stages. STRONG EVIDENCE
- **No invasion.** The idea of an "Aryan invasion" destroying the cities (Wheeler, 1940s) is **rejected**. There is no destruction layer, and the "massacre" skeletons of Mohenjo-daro come from different periods. ESTABLISHED

6. The script: what is known

Fact	Detail
Corpus size ESTABLISHED	In the ICIT export used here: **5,679 records** (inscribed sides) under **4,075 catalogue numbers** from **77 sites**. Of these, 2,290 records are on seals, 2,290 on tablets, 635 on pottery and 262 on tags and sealings. Mohenjo-daro (1,923) and Harappa (2,717) account for 82%. A few objects come from the Gulf (16), Mesopotamia (11) and Iran (11).
Texts are short ESTABLISHED	Average about 3.5 signs per side, median 3. The longest single side in this corpus has **17 signs**, in three lines (seal M-314, Mohenjo-daro). There is no long continuous text.

Number of signs STRONG EVIDENCE	About 400 to 700, depending on how variants are merged: Mahadevan (1977) 417; Parpola (1994) about 386; the ICIT numbering in this corpus 713. Many are rare: 218 sign numbers occur only once. That is too many for an alphabet (20–40) or a pure syllabary (40–100). The usual conclusion is a **logo-syllabic** system, word signs plus sound signs, like Sumerian cuneiform or Egyptian hieroglyphs.
Most frequent sign ESTABLISHED	Sign 740, the "jar", occurs 1,928 times here, usually at the end of a text in reading order. Its meaning is UNKNOWN.
Direction ESTABLISHED	Mostly right to left (4,263 records here; 215 left to right). The evidence is cramped spacing at the left end and overlapping strokes (B. B. Lal 1966; Mahadevan 1977). Seals are carved in mirror image, so the impression reads right to left.
Numerals STRONG EVIDENCE	Groups of short vertical strokes (1–9 or so) behave like numbers or counters. Whether they are numbers or sound signs is not settled.
Same structure everywhere ESTABLISHED	Sign order transfers between cities (this project, rounds 27 and 29). It is one standardised script, used for about 700 years.
Dholavira "signboard" ESTABLISHED	Ten large signs, once inlaid in wood, found near a gateway at Dholavira. It is the largest inscription known, and is still unread.
Bilingual inscription ESTABLISHED	**None exists.** Every successful decipherment of an ancient script has used a bilingual or known names (Egyptian: the Rosetta Stone and royal names; Linear B: Greek place names; Maya: Landa's alphabet and royal names).
Is it writing a language? STRONG EVIDENCE that it is	Farmer, Sproat & Witzel (2004) argued it was a non-linguistic symbol system. Most specialists reject this. The sign order is highly structured, and the texts behave like those of other early writing. This project found that the entropy statistic once used to "prove" it is language (Rao et al. 2009) cannot separate real scripts from non-linguistic ones on real controls (round 52b). So the question rests on the other arguments.
Writing system type UNKNOWN in detail	At matched text length the corpus is **not** like an alphabet or a syllabary. Its statistics are nearest to Shang oracle-bone inscriptions (round 52). Whether it is logo-syllabic or mainly word signs could not be decided statistically (round 52b).

7. The language: what is known, proposed and unknown

Known

- The Meluhhans spoke a language that Mesopotamians needed an interpreter for (the Shu-ilishu seal). ESTABLISHED
- No word of it is securely known. The cuneiform record gives Mesopotamian words for Meluhhan goods, not Meluhhan words (round 51). Even the name "Meluhha" may be a Mesopotamian rendering, or not a native name at all. ESTABLISHED
- No sign of the Indus script has an established sound or meaning. ESTABLISHED

Strong evidence

- **Probably not Indo-Aryan (Sanskrit's ancestor).** Several independent lines point this way:
  - Indo-Aryan is attested first in the Mitanni treaty (c. 1380 BCE) and the Rigveda (c. 1500–1200 BCE), both after the cities.

    - Steppe ancestry, which accompanies Indo-European languages elsewhere, is absent from the Rakhigarhi genome and enters South Asia in the 2nd millennium BCE.
    - The Rigveda borrows hundreds of words from non-Indo-European languages already spoken in the Punjab (Kuiper 1991 lists about 380).
    - The horse and spoked chariot, central in the Rigveda, are not features of Harappan art.

  This does not prove the negative, but it is the mainstream view. STRONG EVIDENCE

- **The Punjab had at least one non-Indo-European language when the Rigveda was composed.** These are the Rigvedic "substrate" words: *kīnāśa* "ploughman", *lāṅgala* "plough", *ulūkhala* "mortar" and others. STRONG EVIDENCE. In this project these words lean toward Proto-Dravidian rather than Munda, but below the bar for attribution (rounds 39–40). The words show what was spoken in the Punjab around 1500 BCE, not what was written on the seals.
- **Hindi farming vocabulary keeps many words of unknown origin** (Masica 1979: about 31% of the agricultural terms examined). This suggests a lost language, sometimes called "Language X", once spoken in the north. STRONG EVIDENCE for a lost substrate; its link to the Harappans is PROPOSED.

Proposed (none demonstrated)

Candidate	For	Against or open
Dravidian (Proto-Dravidian; family of Tamil, Telugu, Kannada, Malayalam, Brahui)	Brahui is spoken in Balochistan, inside the Harappan area. Possible Dravidian loans in the Rigveda. Parpola's and Mahadevan's rebus readings, such as "fish" (*mīn*) = "star". Shared Dravidian words for local plants and farming.	Brahui may be a later migration rather than a relic (Elfenbein 1987). Reconstructed Proto-Dravidian dates to about 2000–1000 BCE and is placed in peninsular India by many. Rebus readings cannot be tested with present methods: the rebus test failed its own validation on Chinese, so it was never run on Indus (round 36). In the anchor tests, Finnish beat its shuffles more strongly than Dravidian did (round 45).
Indo-Aryan / Sanskrit	The name *Sindhu*, and cultural continuities (some yoga-like postures, bangles, the *svastika*).	Section "strong evidence" above. Every tested Sanskrit key performs like its own shuffles (rounds 46–47, 49, 57, 59); a Sanskrit fit is beaten by Finnish and Basque. *Sindhu* in cuneiform is 1st-millennium (round 51 addendum).
Munda / Austroasiatic ("Para-Munda", Witzel)	Some Rigvedic loanwords have Munda-like prefixes. Munda is spoken in eastern India today.	Mostly eastern; the Punjab evidence is thin. Rounds 39–40 lean away from Munda.
A lost language or several	"Language X" in Hindi farming words. Burushaski and Nihali are isolates in the region. A large trading state may have been multilingual.	Not testable without new texts.

Unknown

- Which language family the script wrote. UNKNOWN
- Whether the whole civilisation spoke one language. UNKNOWN (its size makes several likely)
- Whether Brahmi descends from the Indus script. UNKNOWN. Most specialists derive Brahmi from Aramaic. Atlas-style look-alike pairs are no closer than random (round 61).
- What any sign means, including the "jar", the "fish" and the "man". UNKNOWN

8. Claimed decipherments

More than a hundred decipherments have been published (Possehl 1996 reviews many). None is accepted. None has produced a reading that is tested on texts not used to build it and that beats chance. This project tested the ones that could be scored:

Claim	Language	Result here
Yajnadevam (2024)	Sanskrit	Covers the corpus, but its output is no more Sanskrit-like than its own shuffles (z −0.35); a 15-second refit to Finnish or Basque beats it (rounds 46–47). Place-name readings at chance (57); trade words VOID (59).
Rao (1982)	Indo-Aryan (proto-Brahmi values)	NULL (round 49)
Fairservis (1992)	Dravidian	NULL (round 49)
Parpola (1994), Mahadevan (2014), Kak (1988)	Dravidian / Indo-Aryan	Not testable: too few signs given values to score on this corpus (round 49).
Bonta (2023)	Dravidian	Pending: no open copy found.

9. What this project has tested (65 rounds, October 2026)

Every test was registered in advance, run against a known script as a positive control (usually Linear B, which encodes Greek) and against an impossible language as a negative control (Finnish, Basque). The results that survived the controls:

1. **Statistics alone cannot recover sounds** at this corpus size, about 15 uses per sign (round 32). About 40 signs would need to be tied to sounds by outside evidence first.
2. **If the script is logo-syllabic, the corpus is big enough for its sound signs.** The 100 commonest Indus signs average 73 uses, and a Linear B test needed about 29 (round 64).
3. **But a word list of the right language does not work.** Fitting succeeds only with the texts' own words. An independent dictionary of the right language neither decodes nor stands out, even with half the texts' names added (rounds 65, 65b). So fitting a Dravidian or Sanskrit dictionary to the signs cannot settle the language.
4. **Fitting by raw likelihood misleads.** Languages made of open syllables (Hawaiian, Maori, Sanskrit) fit a syllable cipher better than the true language does (round 65). Claims that "language X fits best" are uninformative.
5. **Gulf seals are foreign in sign order** (rounds 30–31).
6. **The Rigvedic substrate leans Proto-Dravidian over Munda**, below the bar for attribution (rounds 39–40).
7. **The script is not alphabet- or syllabary-like** at matched text length; nearest are the Shang oracle bones (round 52).
8. **No family can be attributed** from the proposed Meluhhan loanwords, because none of them is a real Meluhhan word (round 51).

10. What is missing

The missing ingredient is not more statistics. It is outside evidence that ties signs to sounds. The most likely sources:

1. A **bilingual**, for example an Indus inscription with a cuneiform caption, from Mesopotamia, Susa or the Gulf.
2. **Named seals:** a seal, or a sealing on a cuneiform tablet, belonging to a Meluhhan whose name is written in cuneiform. Round 50 searched the open record and found none yet.
3. **More Gulf and Dilmun seals** in good publications, where Indus signs may spell local names.
4. A **longer text**, from new excavation at Indus sites (Ganweriwala and Rakhigarhi are not fully dug).

Arham's aim is to decipher the script, or to prove what is missing and be ready when it appears. The tools in the repository are built to test any such find as soon as it exists.

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- This project: `docs/STATUS.md` and the round reports in `indus-lab/`; corpus counts from `library/inscriptions/inscriptions.csv`.